Chirality Measures and Angular Momentum Generation by Heat Gradient

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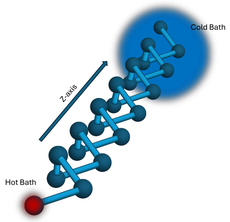
Background: This poster covers two works

- **1.** In a previous work we examined ways to quantify chirality using twisted molecular chains. Here we examine similar properties for small chiral molecules. The key tool we use to measure chirality of molecules is the Continuous Chirality Measure (CCM) introduced by Avnir and his co-workers in early 1990s. The CCM is usually applied to quantify chirality of molecule structure. Now we extend it to characterize the chirality of normal modes of small molecules.

- **2. Angular Momentum Generation by Heat Gradient**

- Chiral phonons are vibrational modes where atoms carries angular momentum, hence carry "handedness" or chirality. It was suggested that such behavior may lead to diode effect in phonon energy transport[1]. It was also suggested that phonon angular momentum can be induced by the heat gradient.

- We study the chiral phonon like behavior in molecule by introducing heat gradient in the helical chain and observe the angular momentum generation in the system.



Chirality Measures

- The expression of the CCM is

$$\mathrm{CCM(Q)} = \min\left\{\frac{1}{2} - \frac{\sum_{i=1}^{N}\langle \boldsymbol{q}_i|\sigma|\boldsymbol{p}_i\rangle}{2\sum_{i=1}^{N}\langle \boldsymbol{q}_i|\boldsymbol{q}_i\rangle}\right\}$$

- Here, $Q = (\boldsymbol{q}_1, \boldsymbol{q}_2, .., \boldsymbol{q}_N)$ represents a molecular configuration defined by the atomic equilibrium position vectors $\boldsymbol{q}_i = (x_i, y_i, z_i)^T$. $P = (\boldsymbol{p}_1, \ldots, \boldsymbol{p}_N) = \mathcal{P}Q$ is a similar molecular configuration obtained from Q by some permutation $\mathcal{P}$ of identical atoms. σ denotes reflection about a given mirror plane and $\langle \boldsymbol{q}_i|\boldsymbol{p}_i\rangle$ is the scalar product of the two vectors $\boldsymbol{q}_i$ and $\boldsymbol{p}_i$. $\mathrm{CCM(Q)}$ is obtained by minimizing the expression on the right over all possible permutations and all possible choices of mirror plane.

- We extend this concept to normal mode CCM by replacing the atomic equilibrium position vectors $\boldsymbol{q}_i$ mass weighted normalized normal mode displacement $\left|\epsilon_{k,i}\right\rangle = (\epsilon_{k,i}^x, \epsilon_{k,i}^y, \epsilon_{k,i}^z)^T$ where $\delta\boldsymbol{q}_i = m_i^{1/2}|\epsilon_{k,i}\rangle$.

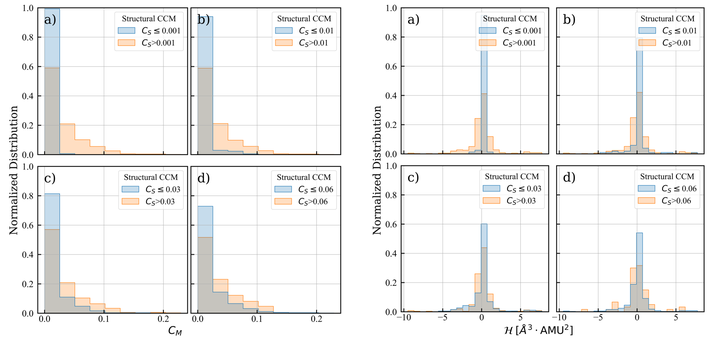
- Another way to evaluate the mode chirality is the momentum pseudo-scalar (mode helicity). It is defined for a given mode k as the dot product of the mode angular momentum and linear momentum along some molecular axis z.

$$\mathcal{H} = \sum_i (p_{k,i}^z L_{k,i}^z)/\dot{A}_k^2$$

1.Chen, H.; Wu, W.; Zhu, J.; Yang, Z.; Gong, W.; Gao, W.; Yang, S. A.; Zhang, L., Chiral Phonon Diode Effect in Chiral Crystals. *Nano Letters* **February 11, 2022**.

Application of Chirality Measures to Small Molecule Set

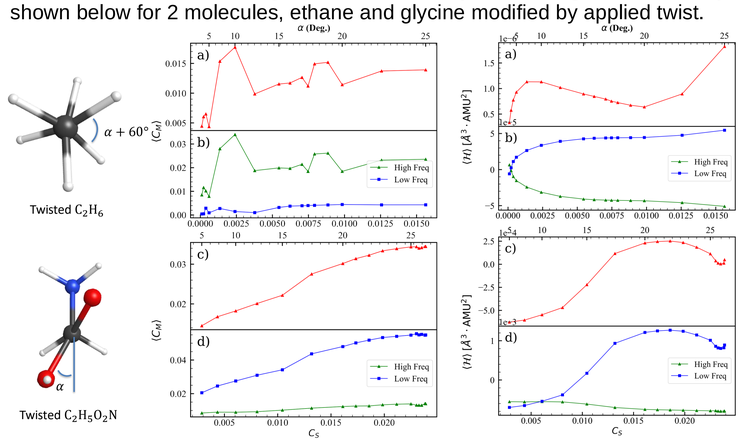
- We applied these chirality measures to small molecule set containing 26 molecules, each molecule includes no more than 14 atoms, mostly C,H,O,N and Si.



- The results displayed above show that normal modes of chiral molecular structures are more likely to appear chiral than those arising from achiral molecular structures. This correlation weakens when a more relaxed structural chirality criterion is used.

Application of Chirality Measures to Twisted Small Molecules

- We have also examined the performance of these chirality measures to twisted versions of some of these small molecules. Twisting was applied by fixing the dihedral angle of certain bonds during relaxation to ground state configuration. The correlation between twist angle, structural chirality and average mode chirality is shown below for 2 molecules, ethane and glycine modified by applied twist.



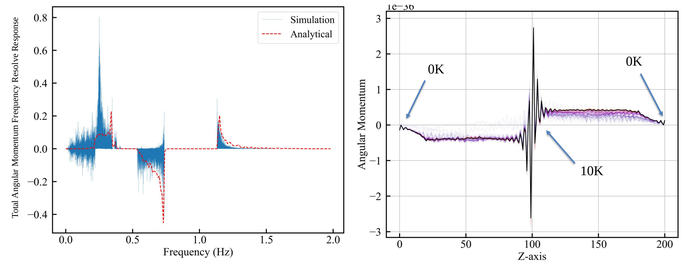
- The average CCM of mode $\langle C_M \rangle$ for both ethane and glycine is very small when $C_S = 0$ and increases as the C_S becomes larger.
- Similarly, the average mode helicity $\langle \mathcal{H} \rangle$ is close to zero for $C_S = 0$ and increases as CS becomes larger.
- The correlation between $\langle C_M \rangle$ and C_S shows opposite trends for low and high frequency modes, with the C_M of the former trends to more positive values while the latter becomes more negative with increasing C_S.
- The correlation of $\langle C_M \rangle$ and C_S for ethane is weaker than in glycine, both separated and unseparated while trend between $\langle \mathcal{H} \rangle$ and C_S appears stronger in ethane.

Angular Momentum Generation by Heat Gradient

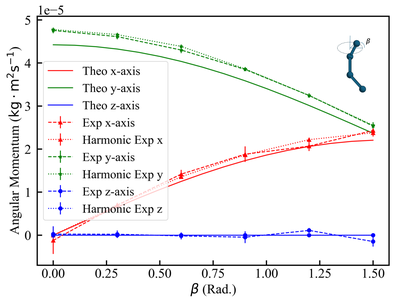
- The system studied is a helical chain constructed as described in Chen's work[1]. It connects between a hot and a cold bath. To save computational time we connect the hot bath to the chain center and cold bath to opposite edges (20 atoms on each side) which make it possible to examine both directions of propagation in one simulation.
- The Hamiltonian of the system is

$$H = \sum_{i=1}^{N} \frac{1}{2} m_i \dot{\boldsymbol{x}}^2 + \sum_{i=1}^{N-1} \frac{1}{2} (\boldsymbol{x}_i - \boldsymbol{x}_{i-1})^T \vec{K}_i (\boldsymbol{x}_i - \boldsymbol{x}_{i-1})$$

- The strength tensor $\vec{K}$ is having orientation preference along the bond.



- We clearly observe the angular momentum generated by the heat gradient and at certain frequency the angular momentum polarization is selected.
- The angular momentum is coupled with the direction of heat flux.
- The total angular momentum induced by thermal gradient is a sum over all atoms and all frequencies. It is strongly frequency dependent and its sign may be inverted in different frequency regimes.



To examine this phenomenon further, we build a 4-atom system which is the smallest unit to have chirality and attach the 2 edges with hot and cold bath to study the angular momentum generation. The interaction potential for the simulation is

$$V = \frac{1}{2} K[|\boldsymbol{x}_i - \boldsymbol{x}_{i-1}| - r_0]^2$$

Z-axis is defined by the central C-C bond.

- At twisted angle equals to zero, the angular momentum is zero along x-axis and z-axis, but non-zero along y-axis.
- The angular momentum along x-axis increases as the twisted angle increases, while the angular momentum along y- axis decreases. The system's symmetry enforced the intensity along x and y axis to equal when the twisted angle β = π/2.
- The angular momentum is always zero along the z-axis.

Acknowledgement

This research is supported by the Air Force Office of Scientific Research under award number FA9550-23-1-0368.